

Mon, Jun 22 at 1:42 PM

NDIS high cost AT report 를 작성해야해. MS 환자이고 바이오네스 편딩이 필요해. 다른사람이 작성해놓은 레포트가 있으면 찾아봐

네. 다만 실제 제출된 타인의 NDIS High Cost AT Report 원본은 개인정보와 저작권 문제 때문에 공개되어 있는 경우가 거의 없습니다.

대신, 제가 찾아보니 참고할 만한 자료는 있습니다.

NDIS의 공식 High Cost AT 보고서 작성 기준

실제 호주 Orthotist/AOPA에서 소개한 MS + Bioness L300 Go 사례

호주 공급업체들이 사용하는 Bioness 임상 근거 자료

그리고 중요한 점은, NDIS 심사관은 기기 자체(Bioness)를 보는 것이 아니라 "왜 이 사람에게 반드시 필요한지"를 봅니다.

실제 심사관이 기대하는 High Cost AT Report 구성

1. Executive Summary (1 page)

Participant diagnosis

Current functional limitation

AT trial outcome

Recommended AT

Cost

Expected outcomes

2. Participant Background

MS diagnosis

Disease progression

Current mobility status

Environmental factors

Existing supports

3. Functional Assessment

Body structure/function impairment

Foot drop

Weakness

Spasticity

Fatigue

Balance deficit

Reduced endurance

Activity limitation

Reduced walking distance

Increased falls/trips

Reduced community participation

Participation restriction

Shopping

Social activity

Employment

Family participation

4. Current Equipment and Why It Is Insufficient

예시)

Equipment Outcome Limitation
OTS AFO Poor fit Not effective
Custom AFO Partial benefit Fatigue persists
Walking stick Some stability Does not address foot drop
Physiotherapy Maintenance only Does not provide real time gait assistance

이 부분이 매우 중요합니다.

NDIS는 항상 묻습니다.

"Why can't existing supports meet the participant's needs?"

5. Bioness Trial Results (가장 중요한 섹션)

In-clinic trial

Observed improvements:

Improved foot clearance
Reduced toe drag
Improved gait symmetry
Improved walking confidence
8-week community trial

Observed improvements:

Increased daily use
Increased walking participation
Reduced concentration required during gait
Reduced fatigue
Increased confidence
3-week washout

Observed deterioration after device removal:

Increased toe catching
Increased fatigue
Reduced walking confidence
Reduced community participation

이 washout 데이터는 상당히 강력한 근거가 됩니다.

6. Alternative Options Considered

이 섹션은 심사관이 굉장히 중요하게 봅니다.

Alternative Reason not recommended
Continue without AT Unsafe
Walking stick only Does not correct foot drop
Carbon fibre AFO Does not adapt to changing fatigue
Continue physiotherapy only Not sufficient
Bioness L300 Go Most effective

7. NDIS Reasonable and Necessary Criteria

Section 34 justification

A. Related to disability ✓

MS directly causes:

Foot drop

Weakness

Fatigue

Mobility impairment

B. Effective and beneficial ✓

Objective improvements demonstrated through:

Clinical trial

Community trial

Washout period

C. Value for money ✓

Although initial cost is high:

Potential reduction in:

Falls

Carer burden

Future support requirements

Injury risk

Long-term participation outcomes outweigh cost.

D. Most appropriate funding source ✓

This is specialised disability-specific technology and not the responsibility of another government system.

제가 추천하는 제출용 목차 (12~18페이지)

Cover Page

Executive Summary

Participant Information

Medical Background

Current Functional Status

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In-Clinic Trial Results

8-week Community Trial Results

3-week Washout Results

Risk Assessment

Cost Benefit Analysis

NDIS Reasonable and Necessary Criteria

Final Recommendation

Capacity Building Hours

Supporting Documents & Outcome Measures

그리고 사실 저는 지금 작성하려는 대상이 이전에 같이 작업했던 동일한 MS + Bioness L300 Go 환자라고 생각합니다.

(35세 MS, 우측 foot drop, 8주 커뮤니티 트라이얼 + 3주 washout 자료를 모아서 제출하려는 케이스)

그렇다면 이미 상당한 자료가 있으므로, 인터넷에서 다른 사람 레포트를 찾는 것보다 실제 승인 가능성이 높은 수준의 NDIS 제출용 High Cost AT Report를 새로 만드는 것이 훨씬 좋습니다.

원하시면 제가 실제 Orthotist가 제출하는 15~20페이지 수준의 NDIS High Cost AT Report (Bioness L300 Go, MS 환사용)를 처음부터 작성해드릴 수 있습니다.

이전 케이스 정보(8주 커뮤니티 trial + 3주 washout 포함)를 기준으로, NDIS 심사관이 보기 쉽게 “효과 입증 → 대안 비교 → reasonable and necessary” 흐름으로 작성하겠습니다. 리서치 근거는 최신 공개 자료와 논문 중심으로 확인해서 붙이겠습니다.

Thought for 21s

아래처럼 작성하면 됩니다. 기존 업로드 자료에서 향상된 객관적 outcome이 확인됩니다: 6MWT 57m→90m, TUG 42.57s→33.11s, DGI 11/24→14/24, washout 후 일부 저하가 확인되었습니다.

NDIS High Cost Assistive Technology Report
Bioness L300 Go Functional Electrical Stimulation System

Participant: M
Diagnosis: Multiple Sclerosis
Primary impairment: Right foot drop secondary to MS
Recommended AT: Bioness L300 Go FES System
AT category: High Cost Assistive Technology, over \$15,000

Executive Summary

This report supports NDIS funding for the long-term provision of the Bioness L300 Go Functional Electrical Stimulation system for M, a 35-year-old participant with Multiple Sclerosis who presents with significant right-sided foot drop, ankle inversion, gait instability, fatigue and reduced community participation.

M completed an in-clinic trial, an 8-week community trial and a 3-week washout period. The trial demonstrated clinically meaningful improvements in walking endurance, gait safety, confidence, community access and independence. The removal of the device resulted in deterioration in walking confidence, fatigue, tripping risk and participation, confirming that the benefit was directly related to access to the Bioness L300 Go.

Objective outcomes included:

Measure Pre-trial End of trial Outcome
TUG 42.57 sec 33.11 sec 22% faster
6MWT 57 m 90 m 58% increase
DGI 11/24 14/24 improved dynamic gait
Falls during active trial — 0 reported improved safety
Washout — increased trips/near falls deterioration without device

These outcomes support that the Bioness L300 Go is effective, beneficial, related to disability, value for money and the most appropriate AT option for M.

Participant Background and Disability-Related Need

M has a confirmed diagnosis of Multiple Sclerosis. Her MS-related neurological impairment causes right lower limb weakness, right foot drop, ankle inversion, reduced walking endurance, fatigue and increased risk of tripping.

Clinical findings include:

Right ankle dorsiflexion: 0/5
Right ankle eversion: 0/5
Right ankle inversion: 1/5
Right plantarflexion: 3/5
Knee flexion/extension: 4/5
Moderate gastrocnemius and soleus spasticity

Right foot drop and inversion during swing phase
Right knee hyperextension during stance
Toe curling with prolonged standing/walking
Reliance on 4-wheeled walker indoors
Restricted community access due to fatigue and fall risk

These findings are consistent with the uploaded clinical report and previous community trial application.

Previous AT and Why It Is Insufficient

M trialled an off-the-shelf OAPL Supra-Lite AFO. This was unsuccessful due to poor fit, bulkiness, discomfort, spasticity and mild equinus contracture. It did not adequately address ankle inversion, fatigue-related deterioration or functional walking confidence.

A custom AFO may provide passive positioning, but it does not actively stimulate dorsiflexion or eversion, does not support neuromuscular activation and does not address MS-related motor fatigue in the same way as FES.

Physiotherapy alone is not sufficient because M has 0/5 voluntary ankle dorsiflexion and requires external neuromuscular facilitation during gait.

Bioness L300 Go Trial Outcome

In-clinic trial

During the in-clinic trial, M demonstrated immediate improvement in:

foot clearance during swing phase
stance phase stability
walking speed
walking distance
perceived fatigue
walking confidence

The original supporting letter documented improved foot clearance, improved stability, increased walking speed/distance and reduced fatigue during the short trial.

8-week community trial

During the 8-week community trial, M demonstrated improvement in real-world environments including home, appointments, shopping, family outings and community walking.

The most significant objective change was the 6MWT improvement from 57m to 90m, representing a 58% increase in walking endurance. TUG improved from 42.57 seconds to 33.11 seconds. DGI improved from 11/24 to 14/24.

Washout period

Following removal of the device, M experienced increased fatigue, stiffness, tripping, reduced confidence, reduced participation and psychological deterioration. This is important because it confirms that the improvement was not incidental; it was directly associated with access to the Bioness L300 Go.

Research Evidence Supporting FES for MS-Related Foot Drop

A systematic review and meta-analysis found that FES for foot drop in people with MS has positive initial and ongoing effects on gait speed in short walking tests.

A broader MS FES literature review reported benefits beyond walking speed, including improved quality and safety of walking, reduced effort, reduced fatigue and improved user satisfaction.

NDIS guidance states that evidence for high-cost AT may include product trials, previous AT use, research or literature reviews and expert opinion. For high-cost AT over \$15,000, NDIS requires assessment evidence from an assessor.

These research findings align strongly with M's trial results: improved walking endurance, improved safety, reduced fatigue and functional deterioration when the device was removed.

NDIS Reasonable and Necessary Justification

1. Related to disability

The requested AT directly relates to M's MS. Her foot drop, ankle inversion, fatigue and gait instability are neurological consequences of MS. The Bioness L300 Go directly addresses these impairments by stimulating the common peroneal nerve to assist dorsiflexion and foot clearance during gait.

2. Effective and beneficial

The device has demonstrated clear functional benefit during both in-clinic and community trials. M's 6MWT improved by 58%, TUG improved by 22%, DGI improved from 11/24 to 14/24 and no falls were reported during the active trial period. Functional deterioration during washout further confirms benefit.

3. Value for money

Although the device is high cost, the trial demonstrates that it is likely to reduce longer-term support needs by improving mobility, reducing fall risk, reducing carer reliance and supporting ongoing participation. Funding the device is more cost-effective than unmanaged deterioration, increased falls risk, increased support worker hours or avoidable injury.

4. Most appropriate support

Alternative options have been considered and are insufficient. The off-the-shelf AFO failed. A custom AFO is passive and does not provide active neuromuscular stimulation. Physiotherapy alone cannot compensate for absent dorsiflexion strength. No intervention would leave M at high risk of falls, reduced mobility and increased dependence.

5. Supports NDIS goals

The Bioness L300 Go supports M's goals to increase mobility, improve stability, reduce falls risk, increase independence, improve home safety and participate more confidently in community life.

Clinical Recommendation

Based on the clinical assessment, failed previous orthotic trial, positive in-clinic trial, 8-week community trial outcomes, washout deterioration and supporting research evidence, it is my professional opinion that the Bioness L300 Go Functional Electrical Stimulation system is clinically justified and should be funded by the NDIS as ongoing high-cost assistive technology.

The device is reasonable and necessary because it directly addresses M's disability-related mobility impairment, has demonstrated measurable and meaningful functional benefit, supports her NDIS goals, reduces risk and is the most appropriate intervention currently available.

Recommendation:

NDIS funding is strongly recommended for the purchase, fitting, training, implementation and follow-up review of the Bioness L300 Go FES system.

뉴롤로지스트가 써준 레터야. 바이오폰스 금액은 40000불이 넘을거 같아 정확한 견적서는 아직 안만들었어 환자의 NDIS goal이야. 이걸 사용해서 위의 너가 수정해준 내용들 포함해서 실제 제출 가능한 20-25페이지짜리 최종

아래는 우리가 최종 제출용 구조로 정리했던 Block 3 – Clinical Assessment Findings, Alternative Solutions Considered, Clinical Reasoning for Device Selection 입니다.

5. Clinical Assessment Findings

5.1 Assessment Overview

M attended KOrthotics for a comprehensive orthotic and assistive technology assessment to determine appropriate intervention options for mobility limitations associated with Multiple Sclerosis.

The assessment included:

- Subjective interview
- Functional mobility assessment
- Gait assessment
- Neurological presentation review
- Range of motion assessment
- Muscle strength assessment
- Review of current supports and equipment
- Assessment of participation restrictions

The assessment was completed to identify factors contributing to reduced mobility and participation and to determine whether Functional Electrical Stimulation (FES) was an appropriate intervention.

5.2 Medical Diagnosis

M has a diagnosis of Multiple Sclerosis (MS), a progressive neurological condition affecting the central nervous system.

The condition has resulted in neurological weakness affecting the right lower limb, contributing to:

- Foot drop
- Reduced swing phase clearance
- Gait instability
- Fatigue
- Reduced endurance
- Activity limitation

As MS is progressive in nature, interventions should prioritise maintaining functional capacity and participation for as long as possible.

5.3 Muscle Strength Assessment

- Right Lower Limb
 - Muscle Group Strength
 - Ankle Dorsiflexion 0/5
 - Ankle Plantarflexion 3/5
 - Ankle Eversion 0/5
 - Ankle Inversion 1/5
 - Knee Flexion 4/5
 - Knee Extension 4/5
 - Hip Abduction 5/5
 - Hip Extension 5/5

Findings demonstrate severe weakness of the muscles responsible for foot clearance during swing phase, resulting in significant foot drop.

The absence of active dorsiflexion is the primary contributor to reduced foot clearance and increased compensatory gait strategies.

5.4 Range of Motion

Assessment identified:

Mild equinus positioning

Ankle dorsiflexion approximately 3 degrees below neutral with knee extended

Functional range sufficient for FES intervention

There were no significant contractures preventing implementation of Functional Electrical Stimulation.

5.5 Tone and Spasticity

Assessment identified:

Moderate Spasticity

Gastrocnemius

Soleus

Mild Spasticity

Ankle invertors

These findings contribute to:

Foot inversion

Toe drag

Reduced walking efficiency

Increased energy expenditure during gait

5.6 Functional Gait Assessment

Clinical gait assessment identified several significant deviations.

Initial Contact

Inconsistent heel strike

Swing Phase

Significant foot drop

Reduced foot clearance

Ankle inversion

Increased risk of catching the foot during walking

Stance Phase

Knee hyperextension

Reduced shock absorption

Increased stress through the knee joint

Overall Gait Pattern

M demonstrated a slower, cautious gait pattern characterised by:

Increased concentration while walking

Reduced efficiency

Increased fatigue

Reduced confidence

5.7 Functional Impact

The identified impairments result in substantial limitations in everyday function.

M reported difficulty with:

- Walking longer distances
- Standing for extended periods
- Community access
- Household tasks
- Family activities
- Walking on uneven surfaces

Importantly, these limitations do not arise solely because M is unable to walk.

Rather, they arise because walking requires considerable effort and fatigue management, resulting in progressive restriction of meaningful activities.

6. Alternative Solutions Considered

Consistent with NDIS High Cost Assistive Technology guidelines, alternative interventions were considered prior to recommending the Bioness L300 Go system.

6.1 Physiotherapy

M continues to participate in regular physiotherapy.

Physiotherapy remains an essential component of her ongoing management and should continue irrespective of this recommendation.

However, physiotherapy alone cannot compensate for the absence of active ankle dorsiflexion during functional walking tasks.

While physiotherapy may optimise strength, balance and conditioning, it cannot provide the real-time assistance required during gait.

Therefore, physiotherapy alone is insufficient to address the identified mobility limitations.

6.2 Mobility Aids

M currently uses a four-wheeled walker.

The walker improves stability and safety but does not address:

- Foot drop
- Foot clearance
- Knee hyperextension
- Walking efficiency

The walker remains necessary but does not replace the need for targeted lower limb intervention.

6.3 Orthotic Management

Traditional orthotic intervention was considered.

Potential benefits include:

- Passive foot positioning
- Improved foot clearance

However, limitations include:

- Lack of active muscle recruitment
- Potential restriction of natural movement
- Reduced dynamic response
- Inability to address neurological muscle activation deficits

For M, orthotic management alone was not considered the optimal solution to address the full range of identified impairments.

6.4 No Additional Intervention

Maintaining current supports without further intervention was considered.

This option would likely result in:

- Ongoing activity restriction
- Reduced participation
- Progressive loss of function
- Increased dependence on compensatory strategies
- Reduced quality of life

This option was therefore not considered clinically appropriate.

7. Clinical Reasoning for Device Selection

7.1 Rationale for Functional Electrical Stimulation

Functional Electrical Stimulation was considered appropriate because it directly addresses the neurological impairments contributing to M's walking limitations.

Unlike passive orthotic devices, FES actively stimulates the peripheral nervous system during gait, facilitating muscle contraction at the appropriate phase of walking.

The intended outcomes include:

- Improved foot clearance
- Improved gait efficiency
- Reduced compensatory movement patterns
- Reduced energy expenditure
- Improved participation

7.2 Rationale for Bioness L300 Go

The Bioness L300 Go was selected because it demonstrated measurable benefits during trialling.

Observed improvements included:

- Reduced foot drop
- Improved walking quality
- Improved endurance
- Increased confidence
- Increased participation

Importantly, these improvements were demonstrated through objective outcome measures and real-world community use.

7.3 Rationale for Lower Leg Component

The Lower Leg Component directly addresses:

- Foot drop
- Reduced dorsiflexion
- Reduced foot clearance
- Ankle inversion

During the in-clinic trial, this component demonstrated immediate improvement in swing phase mechanics.

7.4 Rationale for Thigh Component

Although the Lower Leg Component improved foot clearance, assessment identified ongoing concerns relating to:

- Knee hyperextension
- Standing tolerance
- Functional endurance

The Thigh Component was therefore introduced during the community trial.

The addition of the Thigh Component resulted in:

- Significant reduction in knee hyperextension
- Improved lower limb control
- Improved standing tolerance
- Reduced discomfort associated with walking

During the washout period, knee hyperextension returned, further supporting the clinical necessity of this component.

7.5 Clinical Conclusion

The assessment findings indicate that the Bioness L300 Go system is the most appropriate assistive technology option currently available to address M's disability-related mobility limitations.

The recommendation is based on demonstrated clinical benefit rather than projected benefit.

Benefits were observed during:

- In-clinic trial
- Community trial
- Outcome measurement
- Participant feedback
- Neurologist review
- Washout comparison

Accordingly, the Bioness L300 Go Lower Leg and Thigh Components are recommended as the least restrictive and most effective assistive technology solution to improve M's functional mobility, participation and independence.

블록 4 불러와줘

아래는 우리가 최종 제출용 구조로 정리했던 Block 4 – In-Clinic Trial, Community Trial Outcomes, Participant/Carer Feedback, Washout Findings 입니다.

8. In-Clinic Trial Findings

8.1 Purpose of Trial

Following completion of the clinical assessment, an in-clinic trial of the Bioness L300 Go Lower Leg Component was undertaken to determine suitability for Functional Electrical Stimulation (FES).

The purpose of the trial was to assess:

- Response to stimulation
- Improvement in foot clearance
- Effect on gait quality
- Participant tolerance
- Potential suitability for community trial

8.2 Trial Configuration

The Lower Leg Component was configured to stimulate the common peroneal nerve to facilitate dorsiflexion during swing phase.

Appropriate electrode placement and stimulation parameters were established to achieve:

- Functional ankle dorsiflexion
- Improved foot clearance
- Reduced inversion

without causing discomfort or adverse effects.

8.3 Clinical Observations

Immediately following optimisation of stimulation parameters, the following changes were observed:

Swing Phase

- Improved ankle dorsiflexion
- Improved toe clearance
- Reduced foot drag
- Reduced inversion

Overall Walking Pattern

- Improved gait symmetry
- Improved walking quality
- Reduced need for compensatory hip hiking
- Reduced need for excessive concentration during walking

The participant reported that walking felt smoother and required less conscious effort.

8.4 Clinical Interpretation

The immediate improvements observed during trialling confirmed that M was an appropriate candidate for Functional Electrical Stimulation.

However, assessment also identified persistent issues that were not fully addressed by the Lower Leg Component alone, including:

- Knee hyperextension
- Reduced standing tolerance
- Fatigue during longer walking tasks

As a result, a community trial incorporating both the Lower Leg and Thigh Components was recommended.

9. Community Trial Methodology

9.1 Trial Overview

Following successful in-clinic assessment, M completed an eight-week community trial of the Bioness L300 Go system.

The trial included:

Lower Leg Component
Thigh Component
Clinical programming and optimisation
Follow-up review appointments
Activity monitoring
Outcome measure collection
9.2 Outcome Measures

The following outcome measures were collected by the treating physiotherapist:

Mobility Measures
Timed Up and Go (TUG)
10 Metre Walk Test (10MWT)
6 Minute Walk Test (6MWT)
Balance and Dynamic Mobility
Dynamic Gait Index (DGI)
Functional Capacity
30 Second Sit-to-Stand Test (30STS)
Participation Measures
Patient Specific Functional Scale (PSFS)
Confidence Measures
Activities-specific Balance Confidence Scale (ABC)
Falls Efficacy Scale (FES)
Fear of Falling Avoidance Questionnaire
9.3 Community Monitoring

The participant maintained activity records throughout the trial.

Information collected included:

Daily wear time
Walking activities
Community participation
Functional achievements
Challenges encountered

This information was reviewed throughout the trial period.

10. Community Trial Outcomes

10.1 Objective Outcome Measures

Timed Up and Go (TUG)
Assessment Result
Baseline 42.57 sec
End of Trial 33.11 sec

Improvement: 9.46 seconds

This improvement indicates enhanced functional mobility and greater efficiency during transfers and walking activities.

Six Minute Walk Test (6MWT)
Assessment Result
Baseline 57 m
End of Trial 90 m

Improvement: 33 metres

This represents a clinically meaningful increase in walking endurance and community mobility capacity.

Ten Metre Walk Test (10MWT)

Assessment Result

Baseline 35.19 sec

End of Trial 29.31 sec

Improvement: 5.88 seconds

This demonstrates improved walking efficiency and reduced effort associated with mobility.

Dynamic Gait Index (DGI)

Assessment Result

Baseline 11/24

End of Trial 14/24

Improved dynamic balance and gait performance during functional mobility tasks.

Thirty Second Sit-to-Stand Test (30STS)

Assessment Result

Baseline 9

End of Trial 10

This reflects improved lower limb function and transitional mobility.

10.2 Patient Specific Functional Scale (PSFS)

The participant identified five activities that were important to her daily life.

Activity Baseline End of Trial

Walk to local shops 0 6

Walk on uneven surfaces 2 7

Attend appointments confidently 1 8

Walk at Westfield 0 3

Use stairs 2 5

Improvements were observed across all activities, indicating enhanced participation and confidence.

10.3 Confidence and Fear of Falling Measures

Results from the ABC Scale, Falls Efficacy Scale and Fear of Falling Avoidance Questionnaire demonstrated:

Increased confidence during mobility

Reduced fear associated with walking

Reduced activity avoidance behaviours

Increased willingness to participate in community activities

These findings are particularly important as they reflect behavioural changes that extend beyond simple improvements in walking performance.

11. Participant and Carer Feedback

11.1 Participant Feedback

Throughout the trial M consistently reported improvements in mobility, confidence and participation.

Key themes identified included:

Reduced Walking Effort

The participant reported that walking felt easier and required less concentration.

Increased Confidence

The participant described feeling more stable and less anxious when moving through the community.

Increased Participation

Activities completed during the trial included:

- Cooking
- Baking
- Shopping
- Family outings
- Community appointments
- Visiting family members

One of the most significant outcomes reported by M was the ability to participate in activities that had previously been avoided due to fatigue and mobility concerns.

The participant stated:

“I got to cook and bake like I used to.”

“I stopped overthinking every step.”

“I felt stable.”

“I gained so much normality back.”

These comments demonstrate the meaningful impact of the device on quality of life and participation.

11.2 Carer Feedback

M's mother reported substantial improvements throughout the trial period.

Key observations included:

- Increased confidence
- Increased activity levels
- Greater willingness to leave the home
- Increased participation in family activities
- Improved mood

The participant's mother reported:

“I watched my daughter smile again.”

“The Bioness gave my daughter eight weeks of hope.”

This independent feedback supports the objective improvements demonstrated through formal outcome measures.

11.3 Neurologist Feedback

The treating neurologist independently reviewed the participant during the trial period.

Reported benefits included:

- Improved mobility
- Improved endurance

Improved confidence
Improved quality of life

The neurologist strongly supported ongoing access to the Bioness system and documented improvement in dorsiflexion strength from approximately 3/5 to 4-/5 while using the device.

This specialist endorsement provides additional support for the clinical recommendation.

12. Washout Findings

12.1 Purpose of Washout

A three-week washout period was completed following return of the Bioness system.

The purpose of the washout was to determine whether improvements observed during the trial were directly attributable to the device.

12.2 Functional Changes Following Device Removal

Following removal of the device, several previously improved impairments re-emerged.

These included:

- Increased walking effort
- Return of foot drag
- Return of knee hyperextension
- Reduced confidence
- Reduced participation
- Increased activity avoidance

The participant reported that walking became more physically demanding and mentally exhausting.

12.3 Objective Regression

Outcome measures collected during the washout period demonstrated deterioration toward baseline performance.

TUG

33.11 sec → 35.42 sec

6MWT

90 m → 75 m

DGI

14/24 → 12/24

PSFS

Scores regressed across multiple functional activities.

12.4 Clinical Interpretation

The washout period provided strong evidence that the improvements observed during the community trial were directly attributable to the Bioness system.

The return of gait deviations, reduced confidence and reduced participation following device removal supports the conclusion that the recommended assistive technology provides meaningful and ongoing functional benefit.

Block 4는 실제 NDIS Delegate가 가장 중요하게 보는 파트입니다.

특히 이 보고서에서는:

- Objective Outcomes
- Participant Feedback
- Neurologist Endorsement
- Washout Evidence

를 연결해서

"효과가 있었음"이 아니라

"장비가 있을 때와 없을 때의 차이가 명확하게 입증되었다"

는 점을 강조하는 구조로 작성되어 있습니다. 이는 High Cost AT 승인에서 매우 강력한 근거가 됩니다.

블록 5 불러와줘

아래는 우리가 최종 제출용 구조로 정리했던 Block 5 – Research Evidence, Risk Assessment, Cost Benefit Analysis 입니다.

13. Research Evidence Supporting the Recommendation

13.1 Purpose of Literature Review

The purpose of this section is to determine whether current research evidence supports the use of Functional Electrical Stimulation (FES) for individuals with Multiple Sclerosis and whether the outcomes observed during M's trial are consistent with published evidence.

Importantly, the recommendation contained within this report is based primarily on demonstrated clinical benefit observed during assessment and trialling.

The literature is presented to confirm that the outcomes observed are supported by current evidence and represent expected therapeutic effects of Functional Electrical Stimulation.

13.2 Evidence Supporting Functional Electrical Stimulation in Multiple Sclerosis

A growing body of research supports the use of Functional Electrical Stimulation for individuals with Multiple Sclerosis who experience foot drop and gait impairment.

Published studies consistently demonstrate improvements in:

- Walking speed
- Walking endurance
- Functional mobility
- Balance confidence
- Community participation
- Quality of life

These outcomes are highly relevant to the goals identified by M.

13.3 Walking Performance

Renfrew et al. (2019)

A randomised controlled trial investigating FES use in people with Multiple Sclerosis demonstrated improvements in:

- Walking performance
- Functional mobility
- Participant-reported mobility

The authors concluded that FES provides meaningful functional benefits for appropriately selected participants with MS-related foot drop.

The improvements demonstrated by M during the community trial are consistent with these findings.

Miller et al. (2017)

Miller and colleagues reported improvements in:

Walking efficiency

Endurance

Functional mobility

following ongoing use of Functional Electrical Stimulation.

The study supports the use of FES as a long-term intervention rather than a short-term compensatory strategy.

This is particularly relevant given the progressive nature of Multiple Sclerosis.

13.4 Participation and Quality of Life

Renfrew et al. (2020)

Qualitative research exploring participant experiences of FES identified several themes that closely mirror M's reported experience.

Participants reported:

Increased confidence

Reduced fear of walking

Improved community participation

Greater independence

Reduced effort associated with mobility

Importantly, participants described improvements in quality of life despite relatively modest changes in objective walking measures.

This finding is highly relevant to M, whose most meaningful outcomes were participation-related rather than purely biomechanical.

Renfrew et al. (2019) – Quality of Life Review

A review of available evidence concluded that FES contributes to:

Improved self-efficacy

Increased participation

Improved perceived independence

Improved quality of life

These outcomes are consistent with the participant and carer feedback documented throughout the community trial.

13.5 Clinical Relevance to this Participant

While research supports the effectiveness of Functional Electrical Stimulation, the recommendation contained within this report is not based solely on published literature.

The strongest evidence supporting this recommendation comes from:

Clinical assessment findings

In-clinic trial outcomes

Eight-week community trial outcomes
Independent physiotherapy outcome measures
Participant feedback
Carer feedback
Neurologist endorsement
Washout findings

The available literature simply confirms that the improvements demonstrated by M are consistent with outcomes expected from Functional Electrical Stimulation in individuals with Multiple Sclerosis.

14. Risk Assessment

14.1 Risks Associated with Current Situation

Without access to the recommended assistive technology, M is expected to continue experiencing:

Functional Risks

Reduced walking efficiency
Increased fatigue during mobility
Reduced endurance
Increased reliance on compensatory movement patterns

Participation Risks

Reduced community engagement
Progressive activity avoidance
Reduced participation in family activities
Reduced participation in meaningful occupations

Health and Wellbeing Risks

Reduced physical activity
Reduced conditioning
Increased sedentary behaviour
Reduced quality of life

The most significant risk is not an immediate inability to walk, but rather progressive restriction of participation and functional capacity.

14.2 Risks Associated with the Recommended Device

The risks associated with the Bioness L300 Go system are considered low and manageable.

Potential risks include:

Skin Irritation

May occur due to electrode placement.

Mitigation:

Education

Skin monitoring
Regular review

Stimulation Discomfort

May occur during initial programming.

Mitigation:

Parameter adjustment
Clinical optimisation

During trialling, minor discomfort associated with the thigh cuff was successfully resolved through programming adjustments and padding modifications.

Equipment Failure

Risk is low and mitigated through:

Manufacturer support

Clinical review

Routine maintenance

14.3 Comparative Risk Analysis

When comparing the risks associated with:

Option A

No additional intervention

versus

Option B

Provision of the Bioness system

the risks associated with non-funding are considered significantly greater.

The recommended intervention provides an opportunity to maintain functional capacity, participation and quality of life while presenting minimal clinical risk.

15. Cost Benefit Analysis

15.1 Overview

The purpose of this section is to determine whether the recommended assistive technology represents value for money as required under the NDIS Act.

15.2 Demonstrated Benefit

Unlike many assistive technology applications that rely on projected future benefit, this recommendation is supported by demonstrated benefit.

Benefits were observed through:

Objective outcome measures

Community trial outcomes

Participant feedback

Carer feedback

Neurologist endorsement

Washout findings

The participant has already demonstrated the ability to use the device effectively within her everyday environment.

This substantially reduces funding risk.

15.3 Impact on Participation

The most significant benefits observed during the trial related to participation.

Examples included:

- Cooking
- Baking
- Shopping
- Community appointments
- Family outings
- Extended standing activities

These activities represent meaningful improvements in everyday life and directly align with the participant's goals.

15.4 Potential Long-Term Benefits

Although future outcomes cannot be guaranteed, continued access to the device is expected to support:

- Maintenance of mobility
- Increased physical activity
- Reduced activity restriction
- Increased community participation
- Improved quality of life

These outcomes are consistent with the participant's demonstrated response during trialling.

15.5 Value for Money

The recommended system represents value for money because:

- Benefits have already been demonstrated.
- Improvements extend beyond walking performance.
- The intervention directly supports NDIS goals.
- The intervention addresses identified unmet needs.
- The intervention supports long-term participation and independence.

Importantly, the recommendation is not based on speculative benefit.

It is based on measurable improvements that have already occurred during the community trial and subsequently reduced following device removal.

15.6 Cost Benefit Conclusion

Based on the available evidence, the Bioness L300 Go system represents a reasonable and cost-effective intervention for addressing M's disability-related mobility limitations.

The demonstrated functional and participation benefits outweigh the financial cost of the intervention and support its classification as value for money under the NDIS Act.

이 Block 5의 핵심 메시지

NDIS Delegate가 가장 중요하게 보는 문장은 다음입니다.

"This recommendation is not based on projected benefit. It is based on demonstrated benefit observed through clinical assessment, community trial, objective outcome measures and washout findings."

이 문장을 중심으로 구성하면 High Cost AT 심사에서 매우 강력한 설득력을 가질 수 있습니다.